

MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 45

Level 3 — Advanced · Logical Reasoning I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 3 — Advanced	Topic Focus	Logical Reasoning I — Advanced Series & Coding
Questions	20	Time Allowed	40 minutes
Total Marks	20	Worksheet No.	45 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 45 raises reasoning to Advanced: triangular and Fibonacci-style series, square- and cube-number patterns and odd-ones-out, variable-shift codes (P+1, E+2, N+3), two-step 'number machines' run forwards AND backwards, and ratio analogies. Recognising squares and cubes on sight is the new superpower this sheet builds.

Learning Objectives

- Identify triangular, Fibonacci, doubling, square and cube patterns.
- Decode variable-shift and position codes.
- Run two-step function machines forwards and in reverse.
- Solve ratio and power analogies (8:64, 3:27, 20:5).
- Spot non-squares/non-cubes as odd ones out.

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.



SECTION A · Multiple Choice Questions · 20 Questions

Q1. Find the next number: **1, 3, 6, 10, 15, ____**

- (A) 18 (B) 20
(C) 21 (D) 24

Q2. Find the next number: **2, 3, 5, 8, 13, ____**

- (A) 18 (B) 20
(C) 21 (D) 26

Q3. Using A=1, B=2, C=3 ..., the letters of **CAT** add up to $3+1+20 = 24$. What is the letter sum for **DOG**?

- (A) 26 (B) 24
(C) 20 (D) 18

Q4. In a code, **FACE** is written as **HCEG** (each letter +2). How is **BEAD** written?

- (A) DGCF (B) DGCE
(C) CFBE (D) DFCF

Q5. Complete the analogy: **8 : 64 :: 9 : ?**

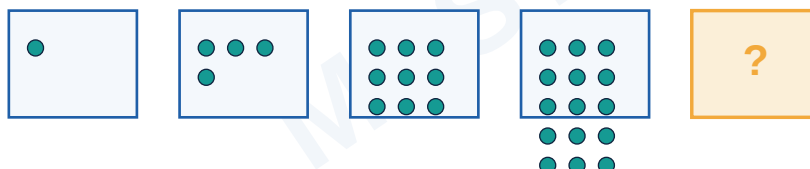
- (A) 72 (B) 81
(C) 90 (D) 99

Q6. Complete the analogy: **3 : 27 :: 4 : ?**

- (A) 48 (B) 64
(C) 16 (D) 36

Q7. Find the missing number: **7, 14, 28, 56, ____**

- (A) 84 (B) 98
(C) 112 (D) 102

Q8. Figure series: count the dots. **1, 4, 9, 16, ?**

- (A) 20 dots (B) 24 dots
(C) 25 dots (D) 30 dots

Q9. If **MONDAY** is the 1st day, **TUESDAY** the 2nd, and so on, what is the 100th day in this weekly cycle?

- (A) Monday (B) Tuesday
(C) Wednesday (D) Sunday



Q10. In a certain code, **1 = A, 2 = B ... 26 = Z**. The word coded as **3-1-20** is:

- (A) CAT (B) DOG
(C) BAT (D) RAT

Q11. Find the odd one out: **8, 27, 64, 100, 125**

- (A) 27 (B) 64
(C) 100 (D) 125

Q12. A number machine does: **$\times 3$, then $- 4$** . If 5 goes in, what comes out?

- (A) 11 (B) 12
(C) 15 (D) 19

Q13. The same machine ($\times 3$, then $- 4$) gives an OUTPUT of 20. What was the INPUT?

- (A) 6 (B) 7
(C) 8 (D) 9

Q14. Find the next term: **Z, X, V, T, ____**

- (A) S (B) R
(C) Q (D) P

Q15. Complete the analogy: **20 : 5 :: 36 : ?**

- (A) 6 (B) 8
(C) 9 (D) 12

Q16. Find the missing number: **100, 81, 64, 49, ____**

- (A) 36 (B) 40
(C) 42 (D) 25

Q17. In a code, **PEN** becomes **QGQ** ($P+1, E+2, N+3$). How does **CAR** become?

- (A) DCU (B) DBU
(C) DCT (D) DBT

Q18. Find the next number: **3, 6, 12, 24, 48, ____**

- (A) 72 (B) 84
(C) 96 (D) 60

Q19. A machine adds 7 to a number, then halves the result. If 9 goes in, what comes out?

- (A) 8 (B) 7
(C) 9 (D) 16

Q20. Find the odd one out: **121, 144, 169, 180, 196**

- (A) 121 (B) 144
(C) 180 (D) 196

ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	C	A	A	B	B	C	C	B	A
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	A	C	B	C	A	A	C	A	C

Step-by-Step Solutions

Q1. Correct answer: (C)

These are triangular numbers; gaps grow +2, +3, +4, +5, +6: $15 + 6 = 21$.

Q2. Correct answer: (C)

Each term is the SUM of the two before it (Fibonacci-style): $8 + 13 = 21$.

Q3. Correct answer: (A)

D=4, O=15, G=7. Sum = $4 + 15 + 7 = 26$.

Q4. Correct answer: (A)

Each letter moves +2: B→D, E→G, A→C, D→F → **DGCF**.

Q5. Correct answer: (B)

$8 \times 8 = 64$ (the number squared). So $9 \times 9 = 81$.

Q6. Correct answer: (B)

$3 \times 3 \times 3 = 27$ (cubed). So $4 \times 4 \times 4 = 64$.

Q7. Correct answer: (C)

Each number is DOUBLED: $56 \times 2 = 112$.

Q8. Correct answer: (C)

Square numbers: 1, 4, 9, 16, then $5^2 = 25$ dots.

Q9. Correct answer: (B)

$100 \div 7 = 14$ remainder 2. Remainder 2 means the 2nd day of the cycle: **Tuesday**.

Q10. Correct answer: (A)

3 = C, 1 = A, 20 = T → **CAT**.

Q11. Correct answer: (C)

$8 = 2^3$, $27 = 3^3$, $64 = 4^3$, $125 = 5^3$ — all perfect CUBES. **100** is a square (10^2), not a cube.

Q12. Correct answer: (A)

$5 \times 3 = 15$; $15 - 4 = 11$.

Q13. Correct answer: (C)

Work backwards: $20 + 4 = 24$; $24 \div 3 = 8$.

Q14. Correct answer: (B)

Letters go backward skipping one: Z, (Y), X, (W), V, (U), T, (S), **R**.

Q15. Correct answer: (C)

$20 \div 4 = 5$ (divide by 4). So $36 \div 4 = 9$.

**Q16. Correct answer: (A)**

Square numbers going down: 10^2 , 9^2 , 8^2 , 7^2 , so next is $6^2 = 36$.

Q17. Correct answer: (A)

$C+1 = D$, $A+2 = C$, $R+3 = U \rightarrow \mathbf{DCU}$.

Q18. Correct answer: (C)

Each number is doubled: $48 \times 2 = \mathbf{96}$.

Q19. Correct answer: (A)

$9 + 7 = 16$; $16 \div 2 = \mathbf{8}$.

Q20. Correct answer: (C)

$121 = 11^2$, $144 = 12^2$, $169 = 13^2$, $196 = 14^2$ — all squares. **180** is not a perfect square.

